ECE 510 SPRING 2026

CODEFEST - 2

CMAN - 2

TASKS:

Peak Compute = 10TFLOPS (FP32) peak compute
= 10,000 GFLOPS

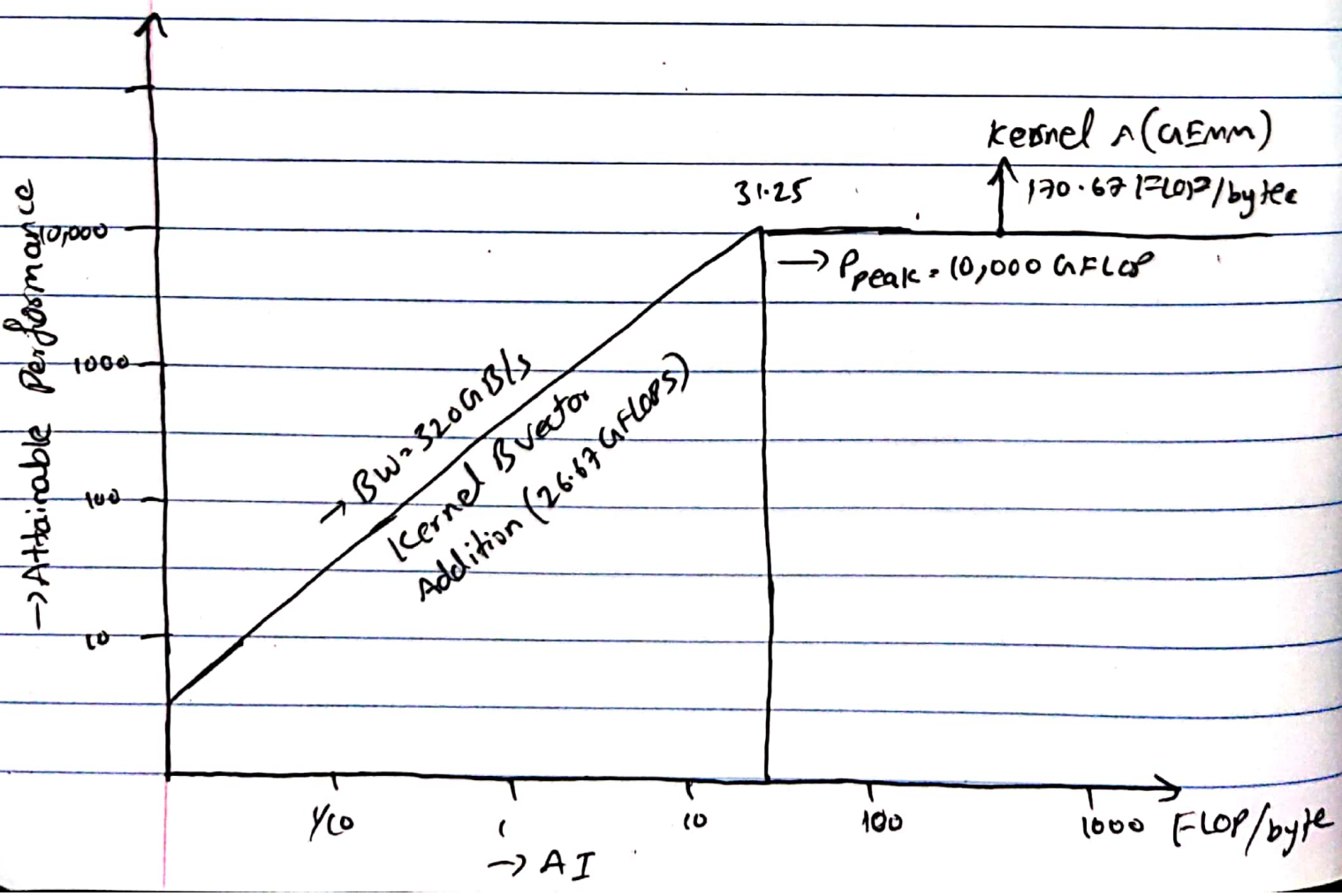
DRAM Bandwidth = 320 GB/s

Ridgepoint = 10,000/320
= 31.25 FLOP/Byte

1. P = Arithmatic Intensity x B

P = Ppeak = 10,000 GFLOPS

Ridge point = (31.25 FLOP/Byte)



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2. Kernel A - Dense GEMM (two matrix 1024x1024)

FLOPS = $2 \times N^3$

$= 2 \times (1024)^3$

$= 2,147,483,648$ FLOPs

Bytes = One Matrix = $1024 \times 1024 \times 4 = 4,194,304$ Bytes

~~6~~, 3 Matrices $\times$ 4,194,304

$\Rightarrow$ 12,582,912 Bytes

Arithmatic Intensity (AI)

$$AI = \frac{FLOPS}{Bytes} = \frac{2147,483,648}{12,582,912} = 170.67 \text{ FLOP/Bytes}$$

= 170.67 FLOP/Bytes

this is > 31.25 (compute bound is above)

Attainable performance = 10,000 GFLOPS/s

3. Kernel B - Vector addition:

one matrix = 4,194,304 Bytes (1024 x 1024 x 4)

one vector = 16,777,216 Bytes (4,194,304 x 4)

3 vectors = 3 x 16,777,216

= 50,331,648 Bytes

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Arithmetic Intensity (AI):

$$\frac{4,194,304}{50,331,648} = 0.083 \text{ FLOP/Byte}$$

AI = 0.083 FLOP/Byte

which is < 31.25 (below ridge point which is memory bound).

Attainable performance = 320 × 0.083

= 26.67 G FLOP/s

4. kernel A → Compute Bound

kernel B → Memory Bound.

for kernel A → More T FLOPS

for kernel B → More Mem Bandwidth.

for level A, increase the Arithmetic units so that the problem can operate at much higher peak performance.

∵ Since kernel A is operating on the ridge point on right side, it has enough data to work on, but not enough compute units.

→ For kernel B, the AI is 1/12 so it needs very high bandwidth memory, using wider data BUS, HBM and operating at higher freq DRAM would improve the utilization.

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